

Peter Mao

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Education

Drexel University Ph.D. in Mathematics (expected 2027) Advisor: Darij Grinberg M.S. in Mathematics, 2024	2022 – present
Cornell University B.A. in Mathematics, <i>cum laude</i> , 2022	2018 – 2022

Research Interests

Algebraic combinatorics, dynamical algebraic combinatorics,
symmetric group algebras, permutation statistics.

Publications and Preprints

1. *Foatic homomesy*. Preprint (2026). [PDF](#).
2. (with Darij Grinberg) *Necklaces over a group with identity product*. *Journal of Integer Sequences* **29** (2026), Article 26.4.6. [Journal version](#). [arXiv:2405.08937](#).

Talks

- *Foatic homomesy: telescoping proofs for orbit-averages on S_n* . May 2026
SIAM Student Chapter, Drexel University
- *Necklaces over a group with identity product: counting and bijective connections*. November 2024
SIAM Student Chapter, Drexel University

Conferences and Workshops

- Mid-Atlantic Algebra, Geometry, and Combinatorics Workshop (MAAGC), October 2026
Virginia Commonwealth University (*upcoming*)
- Formal Power Series and Algebraic Combinatorics (FPSAC 2026), July 2026
University of Washington
- Triangle Lectures in Combinatorics, Wake Forest University April 2026

Teaching

Instructor of record, Drexel University

- MATH 122, Calculus II Fall 2026
- MATH 221, Discrete Mathematics (mastery-based testing redesign) Summer 2026
- MATH 101, Introduction to Analysis I Fall 2024
- MATH 221, Discrete Mathematics Summer 2024

Teaching assistant, Drexel University

- MATH 311, Probability and Statistics I Winter 2024, Spring 2025, Spring 2026
- MATH 200, Multivariable Calculus Spring 2024, Winter 2025, Fall 2025
- MATH 121, Calculus I Winter 2026
- MATH 102, Introduction to Analysis II Fall 2022 – Spring 2023

Tutor, Math Resource Center, Drexel University Fall 2022 – present

Undergraduate teaching assistant, introductory physics labs, Cornell University Fall 2019

References

Available upon request.